

# Shayan Shakeri

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Toronto, Canada · Canadian citizen · TN-eligible (no sponsorship required)

## PROFESSIONAL SUMMARY

Robotics research engineer working on manipulation, reinforcement learning, and real2sim2real. Co-founder of Merlyn Labs; placed 8th in Stanford's BEHAVIOR-1K Challenge, training manipulation policies on 10,000+ demonstrations across 22 tasks and identifying proprioceptive collapse as a critical VLA failure mode (60% masking improved task success up to 48%). Open-sourced a flow-matching VLA integration for RLinf enabling RL on BEHAVIOR-1K in OmniGibson. Full stack from PCB and sim through real hardware.

## CORE COMPETENCIES

Robot Manipulation & Learning

Reinforcement Learning

Real2Sim2Real / Sim-to-Real

VLA / Robotic Foundation Models

Imitation Learning

Multimodal Sensor Fusion

## WORK EXPERIENCE

**Merlyn Labs** Aug 2025 - Present

**AI Researcher & Co-Founder** | Toronto, Canada

- Open-sourced a **flow-matching VLA integration for RLinf**, enabling **RL for manipulation** on the BEHAVIOR-1K suite in OmniGibson.
- Running **sim-to-real (real2sim2real) transfer** of household manipulation tasks on a hand-built AlohaMini embodiment.
- Showed  $\pi 0.5$  generalization failures on LIBERO-PRO are recipe-induced overfitting; conservative finetuning doubled position-swap success (21% to 42%), evidence of spatial generalization over memorization.

**BMO AI Centre of Excellence** Sep 2024 - Present

**AI Research Engineer** | Toronto, Canada

- Built a deterministic agent eval pipeline over hundreds of synthesized inputs to detect misaligned outputs at scale.
- Uncovered a systematic bias to downplay investment risk in a GenAI tool serving **\$200B+ AUM** wealth management.
- Building a graph-based agentic system answering complex multi-hop relational queries across bank client data.

**Epineuron Technologies** May 2021 - Aug 2022

**Biomedical & Electrical Engineering Co-op** | Mississauga, Canada

- Designed and assembled PCBs and authored validation protocols (IEC 60601-1, ISO 13485) for PeriPulse, an FDA Breakthrough-designated device now in multi-national clinical trials.

## PROJECTS

**Stanford BEHAVIOR-1K Challenge** 8th Place | Standard Track

Trained manipulation policies on 10,000+ demonstrations across 22 tasks. Identified proprioceptive collapse as a critical VLA failure mode (60% masking improved task success up to 48%); chunked execution outperformed temporal ensembling 3x, revealing missing temporal awareness.

Robot manipulation | RL | Imitation learning | Failure-mode discovery

## EDUCATION

**M.Eng in AI & Robotics** | **University of Toronto** Sep 2024 - Apr 2027 (anticipated)

Coursework: Reinforcement Learning, AI Applications in Robotics, Deep Learning, Healthcare Robotics.

**B.Eng, Engineering Physics (Co-op)** | **McMaster University** Sep 2018 - Jun 2023

Dean's Honour List. Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems.

## SKILLS

**Robotics:** Robot Manipulation, RL for Manipulation, Real2Sim2Real, VLA Models, Behavior Cloning, OmniGibson, MuJoCo, Flow Matching, Sim2Real, ROS2, Grasp Planning

**ML & AI:** Python, C/C++, PyTorch, JAX, RL, Imitation Learning, LLM Fine-tuning, Multimodal Sensor Fusion (YOLO + LiDAR)

**Hardware:** PCB Design, COMSOL, 3D Printing, Embedded Systems